

Fault tolerance: Checkpointing and recovery mechanisms to handle node failures during long-running tasks.

Diverse neural networks: Built-in support for CNNs, RNNs, GANs, and reinforcement learning models.

Customisable layers: Low-level APIs (C++/Python) for fine-grained control, plus high-level APIs (like Singa-Easy) for rapid prototyping.

Dynamic and static graphs: Hybrid computation graph support for both ease of use and optimisation.

Edge and cloud deployment: Lightweight export options for mobile/IoT devices and cloud platforms.

Model compression: Techniques like quantization to reduce model size for resource-constrained environments.

Installing Apache SINGA

First, ensure your system meets the following requirements.

Operating system: Linux (Ubuntu 20.04/22.04 recommended) or macOS.

Dependencies:

- Python 3.6+ and pip
- CMake 3.1+
- GCC 7+ or Clang 5.0+
- OpenMPI 4.0+ (for distributed training)
- CUDA 10.2/11.x and cuDNN 8.x (for GPU support)
- OpenCV (optional, for image processing)
- Protocol Buffers (protobuf)
- BLAS (e.g., OpenBLAS, Intel MKL)

Install dependencies on Ubuntu with the following code:

```
#sudo apt update
#sudo apt install -y build-essential cmake python3-dev
python3-pip \
  libopenblas-dev libopencv-dev protobuf-compiler
libprotobuf-dev \
  openmpi-bin libopenmpi-dev
```

Option 1 to install SINGA: Install via pip (CPU-only).

For a quick CPU-only installation, type:

```
pip install singa
```

Option 2 to install SINGA: Build from source (GPU support).

Clone the repository:

```
git clone https://github.com/apache/singa.git
cd singa
```

Configure the build (enable CUDA, OpenMPI, etc):

```
mkdir build
cd build
```

```
cmake -DCMAKE_INSTALL_PREFIX=/usr/local \
  -DENABLE_CUDA=ON \
  -DENABLE_DIST=ON \
  -DENABLE_TEST=ON \
  ..
```

Build and install using the following code:

```
make -j$(nproc) # Use all CPU cores
sudo make install
```

Install Python bindings:

```
cd ../python
pip install .
```

Configure environment variables as follows:

```
Add the following to your ~/.bashrc or ~/.zshrc:
export SINGA_HOME=/path/to/singa # Path to SINGA source
directory (if built from source)
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/local/lib
export PATH=$PATH:/usr/local/bin
```

Reload the shell:

```
source ~/.bashrc
```

Verifying the installation

Test SINGA in Python as follows:

```
import singa
print(singa.__version__) # Should output the installed
version
```

Test GPU support:

```
import singa
print(singa.gpu_devices()) # List available GPUs
```

Configuring distributed training

For multi-node training, set up MPI. Ensure OpenMPI is installed and nodes can communicate via SSH without passwords.

Create a hostfile (e.g., hostfile) listing all worker nodes:

```
worker1 slots=2 # 2 GPUs on worker1
worker2 slots=1 # 1 GPU on worker2
```

Run a distributed job:

```
mpirun -np 3 -hostfile hostfile python train.py
```